DEVELOPED PROJECT PROPOSAL

Deployable segmentation of brain tumors from MRIs through deep learning

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Written portion (6%)	10-9	8-7	6-5	4-3	2-0	Grade /10	Weighted
Format & English: organisation, spelling, sentence structure, etc.	Writing exceeds expectations. Very few if any mistakes, proper scientific writing	Writing is as expected for this level. Could use some work to improve.	Writing and format needs improvement	Needs significant improvement	Needs a complete re-write	8 /10	4 /5
Introduction contains relevant background information.						9 /10	18 /20
In-text citations utilized properly.						10 /10	15 /15
Introduction uses scientific writing style.						7 /10	3.5 /5
Project CLEARLY defined						9 /10	9 /10
Precise, concrete, progress since preliminary proposal						8 /10	4 /5
						53.5	Total: /60
Project Schedule (2%)	**5**	**4**	**3**	**2**	1-0		
Feasible and realistic goals set.	All elements addressed	Almost all elements present	Some elements missing	Most elements missing	Unclear or undeveloped	5	/5
Includes intermediate and large milestones.						5	/5
Plan to divide work						5	/5
Organized						5	/5
						20	Total: /20
Annotated Bibliography (2%)		4	3	2	1-0		
Reference 1, Annotation & Scholarly		Appropriate well annotated, and formatted properly	At least one formatting error, not using Vancouver style, misleading annotation	More issues needs significant improvement.	References missing or not appropriate or not annotated.	4	/4
Reference 2, Annotation & Scholarly						4	/4
Reference 3, Annotation & Scholarly						4	/4
Reference 4, Annotation						4	/4
Reference 5, Annotation						4	/4
						20	Total: /20

93.5

Disciplines bullet list:

(1) Biology
(2) Computer science

it is a diagnostic assistance tool, not a webapp. this is the place to sound more scientific and less like you are trying to sell your product.

Project type: Web-app design and machine learning algorithm development.

Logistical requirements: Sufficiently powerful compute units (at home) with graphics cards capable of training deep learning models (Bolin's computer - GPU: RTX 4060, Malik's computer – GPU: GTX 1660 Super). Training our models will require continuous operation overnight (estimated 1 week combined runtime).

Integration and context: From a biological perspective, this project builds on concepts learned in the Human Anatomy and Physiology course, particularly in the chapter covering the nervous system. While the course provides a general focus on the central nervous system (CNS) and peripheral nervous system (PNS), this project applies the fundamental principles of brain structure and pathology. A strong understanding of brain anatomy is essential to interpret MRI images accurately, identify abnormal tissue patterns associated with tumors, and assess potential false positives in our analysis. The brain can be divided into major anatomical regions, including the cerebral cortex (part of the cerebrum), the forebrain (which includes the cerebrum, thalamus, and hypothalamus), the midbrain, and the hindbrain (which includes the pons, medulla oblongata, and cerebellum). This knowledge allows us to understand how tumor location can influence neurological function and helps deepen the interpretability of our machine learning model. In addition, understanding cancerous cell proliferation in the brain through abnormal mitosis enables us to physiologically differentiate healthy cells from tumor cells. Furthermore, in enriched courses such as Waves and Modern Physics, students were introduced to the fundamental principles of X-ray imaging and magnetic resonance imaging (MRI). By expanding these concepts and focusing on the behavior of hydrogen nuclei found in water molecules throughout the body, we can understand how radiofrequency pulses applied within a strong magnetic field generate measurable signals. These signals are then used to create detailed images of brain tissue, which form the basis of the data used in our model. Together, these biologically focused concepts establish an essential scientific foundation that will be integrated into our machine learning model for accurate brain tumor detection.

From a computer science perspective, this project applies principles of algorithmic design, data structures, and machine learning using Python, which extends over the learnings in the Programming in Science course. Developing a 3D U-Net requires a solid understanding of tensor operations, optimization algorithms, and loss functions. Building the secure, locally deployable web application relies on web development frameworks including React, Next.js, and Tailwind CSS. These components collectively demonstrate the practical application of computational theory to solve complex medical imaging problems while ensuring data privacy through local containerized deployment [11].

This is a bold DIRECT link that you are implying. There are also possible increases due to better detection tools, more people getting checked, other environmental issues. Careful to not make such bold scientific statements in this type of paper.

Project introduction:

Brain tumors are serious neurological conditions whose causes often remain unknown yet have seen a sudden increase in diagnosis due to an aging population. According to Ostrom QT et al.'s statistical report, "[t]here were 86,030 deaths attributed to malignant brain and other CNS tumors between 2016 and 2020" [1]. Despite advances in neuroimaging, the five-year survival rate for malignant brain tumors remains below 40% [1], highlighting the urgent need for improved and more reliable diagnostic methods. These statistics emphasize the importance of early and accurate detection, particularly through advanced imaging techniques such as magnetic resonance imaging (MRI) and current advances in machine learning (ML) algorithms. The objective of this integrative project is to develop a reliable machine learning model to solely aid doctors and radiologists in improving the efficiency and certainty of brain tumor diagnosis. While artificial intelligence-based tumor detection models have increasingly been explored in medical research, many existing systems remain unavailable for practicing clinicians and limited in terms of their three-dimensional capabilities. By integrating established biological knowledge of brain tumor pathology with advances in three-dimensional image segmentation, specifically using 3D U-Nets, the system will analyze MRI scans and highlight regions with a high probability of tumorous tissues.

get rid of filler words. Just begin, don't introduce that you are beginning.

~~To begin,~~ brain tumors arise from an abnormal proliferation of cells and are broadly classified by the World Health Organization (WHO) as either primary or secondary (metastatic). Among primary brain tumors, gliomas represent the most common malignant subtype and originate from glial cells, which provide structural and metabolic support to neurons. Glioblastomas, also referred to as high-grade gliomas, are the most common and most lethal malignant subtype [1]. Classified as grade IV gliomas, glioblastomas are characterized by aggressive proliferation of cancerous cells, diffuse infiltration into surrounding brain tissue, neuronal death and abnormal formation of blood vessels [6]. These pathological features make complete surgical resection nearly impossible at times. For this integrative project, a dataset containing glioblastoma cases was specifically selected because of the complexity of their diagnosis. This tool will help confirm preliminary diagnoses, surgical planning, radiation targeting and monitoring therapeutic response.

MRIs are still identified as the best solution to identify brain tumors due to their superior soft tissue contrast and multiparametric imaging capabilities. MRI functions by aligning hydrogen protons in the body within a strong magnetic field and detecting the signals emitted as they return to their original orientation. In fact, image contrast is determined by tissue-specific relaxation properties known as T1 (longitudinal relaxation) and T2 (transverse relaxation) times. Different variations of these measurements allow clinicians to visualize different features of glioblastomas. For example, T1-weighted images highlight the formation of abnormal blood vessels in the brain. In addition, T2-FLAIR sequences can focus on the infiltration of the tumor within the interstices of the brain by suppressing the signal coming from the cerebrospinal fluid [2,7]. Because each modality reveals different pathological information regarding the diagnosis

of glioblastomas, we took these parameters into account during our selection of datasets, enabling our ML model to learn from the same multimodal information that physicians rely on. Incorporating these sequences improves segmentation accuracy by differentiating all elements in the brain and limiting the formation of false positives during later testing [5].

To process this multimodal MRI data, our system employs a 3D U-Net architecture, a convolutional neural network designed specifically for biomedical image segmentation [4]. Because clinical environments often face missing scans due to time constraints, we will implement a two-pronged experimental pipeline. First, to establish a reliable baseline, we will train standard modality-specific 3D U-Nets. This approach involves training independent models on individual modalities, such as T1 or FLAIR, as well as a model trained strictly on the complete set. This baseline will allow us to quantify the exact predictive value of each structural scan.

is that a term you invented?

Following this, we will develop a modality-agnostic model using a modality dropout technique. During training, random subsets of the input channels will be masked with zeros [12]. According to Havaei et al., this forces the network to learn robust and shared latent representations rather than overfitting to a specific combination of MRI sequences [12]. By utilizing independent feature encoders for each modality that fuse in a centralized bottleneck, the network can dynamically aggregate available information and perform accurate inference even with incomplete data [12]. This expanded approach bridges the gap between controlled academic datasets and unpredictable clinical realities, serving as a foundational step toward federated learning where heterogeneous data distributions are common.

To address the severe class imbalance inherent to tumor segmentation where healthy tissue overwhelmingly outnumbers tumorous tissue, we will optimize the network using Dice loss. This loss function directly maximizes the overlap between our predictions and the ground truth annotations. Finally, the best performing model will be integrated into a self-hosted web application built with React, Next.js and Tailwind CSS [5]. The entire system will be deployed via Docker to ensure strict data privacy and localized inference, keeping patient data securely on premises [11].

~~In conclusion~~, this integrative project bridges biology and computer science to address a critical need in neuro-oncology. By combining a physiological understanding of glioblastoma pathology with a robust, modality-agnostic 3D U-Net architecture, we aim to deliver a highly accurate and resilient segmentation tool. The deployment of this model through a secure, localized web application ensures that clinical workflows are enhanced without compromising patient data privacy [5, 11]. Finally, this work demonstrates the potential of deep learning to assist medical professionals in providing faster and more reliable brain tumor diagnoses.

there is no conclusion to the introduction to a paper. just say the information

Project Schedule:

Week	Bo Lin (Bio+CS)	Malik (Bio+UI)	Kuan Yi (CS+webapp)	Zehao (CS+webapp)
1 (March 2)	Implement Figma UI prototype; Tumor and MRI research	Implement Figma UI prototype; Tumor and MRI research	Set up PyTorch/CUDA environment; download and explore BraTS dataset	Begin data preprocessing pipeline (normalization, resampling)
2 (March 9)	Finalize Figma prototype; begin React/Next.js project setup and basic routing; train ai model; Tumor and MRI research	Finalize Figma prototype; begin React/Next.js project setup and basic routing; train ai model; Tumor and MRI research	Implement baseline 3D U-Net architecture in PyTorch; launch training run; Tumor and MRI research	Complete skull stripping and preprocessing; feed cleaned data into Kuan Yi's pipeline; Tumor and MRI research
3 (March 16) First Submission	Prepare Presentation (Tumors + web app); connect Model to application; record baseline Dice score/IoU results	Prepare Presentation (MRI + web app); connect Model to Website; compile and review application and presentation; bibliography	Prepare Presentation (U-net); connect Model to Website; Record baseline Dice score/IoU results	Prepare Presentation (preprocessing + U-net); compile and review application and presentation; bibliography
4 (March 23) Presentation	Practice presentation + break	Practice presentation + break	Practice presentation + break	Practice presentation + break
5 (March 30)	Compare performance of different variant	Compare performance of different variant	Build the backend inference pipeline connecting the trained model to the web app	Build the backend inference pipeline connecting the trained model to the web app
6 (April 6) Peer Review	Implement Reviews; finalize model weights and document final performance metrics	Implement Reviews; finalize model weights and document final performance metrics	Implement Reviews; revise biology/analysis sections based on instructor	Implement Reviews; revise biology/analysis sections based on instructor
7 (April 13)	End-to-end testing; Bug fixing	UI polish	UI polish	End-to-end testing; Bug fixing
8 (April 20)	Final Review of ML section	Final Review of Biology section	Final Review of ML section	Final Review of web app
9 (April 27) Final Submission + Team assesment	Submit Final Project	Submit Final Project	Submit Final Project	Submit Final Project

References

[1]

Ostrom QT, Price M, Neff C, Cioffi G, Waite KA, Kruchko C, et al. CBTRUS statistical report: primary brain and other central nervous system tumors diagnosed in the United States in 2016-2020. Neuro Oncol [Internet]. 2023 Oct 4 [cited 2025 Feb 21];25(Suppl 4):iv1–iv99. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC10550277/

Annotation: This report documents incidence, prevalence, and survival rates of primary brain and CNS tumors in the United States from 2016 to 2020, using data from 50 state cancer registries. Key findings show approximately 94,390 annual diagnoses, with glioblastoma as the most lethal subtype and five-year survival below 36%. This source establishes the rising clinical burden that justifies our need for an automated MRI-based detection platform.

[2]

Villanueva-Meyer JE, Mabray MC, Cha S. Current clinical brain tumor imaging. Neurosurgery [Internet]. 2017 May [cited 2025 Feb 21];81(3):397–415. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC5581219/

Annotation: This review examines MRI techniques used clinically for brain tumor diagnosis, covering T1, T2, FLAIR, and diffusion sequences. The authors conclude that MRI interpretation requires specialist expertise and is time-intensive, directly establishing the clinical gap our project addresses by automating image segmentation.

[3]

Ronneberger O, Fischer P, Brox T. U-Net: convolutional networks for biomedical image segmentation. In: Navab N, Hornegger J, Wells W, Frangi A, editors. Medical image computing and computer-assisted intervention – MICCAI 2015 [Internet]. Cham: Springer; 2015 [cited 2025 Feb 21]. p. 234–241. Available from: https://link.springer.com/chapter/10.1007/978-3-319-24574-4_28

Annotation: This paper introduces U-Net, an encoder-decoder convolutional network designed for biomedical image segmentation with limited training data. Skip connections preserve spatial detail critical for tumor boundary detection. Evaluated on two benchmarks, U-Net achieved state-of-the-art accuracy and is the direct architectural foundation of our model.

[4]

Yousef R, Khan S, Gupta G, Siddiqui T, Albahlal BM, Alajlan SA, et al. U-Net-based models towards optimal MR brain image segmentation. Diagnostics (Basel) [Internet]. 2023 May 4 [cited 2025 Feb 21];13(9):1624. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC10178263/

Annotation: This study compares U-Net variants (Attention U-Net, Residual U-Net, Dense U-Net) on brain MRI datasets using Dice score, IoU, sensitivity, and specificity. Attention and Residual U-Net outperform standard U-Net on irregular tumors. This paper provides our evaluation methodology and performance benchmarks.

[5]

Dorfner FJ, Patel JB, Kalpathy-Cramer J, Gerstner ER, Bridge CP. A review of deep learning for brain tumor analysis in MRI. NPJ Precis Oncol [Internet]. 2025 Jan 3 [cited 2025 Feb 21];9:2. Available from: https://www.nature.com/articles/s41698-024-00789-2

Annotation: This review identifies barriers to clinical deployment of deep learning MRI models, including dataset bias, HIPAA restrictions, and cloud infrastructure risks. The authors conclude local inference is the safest pathway, directly validating our architectural decision to prioritize local inference and mandatory radiologist oversight.

[6]

Schaff LR, Mellinghoff IK. Glioblastoma and other primary brain malignancies in adults: a review. JAMA [Internet]. 2023 Feb 21[cited 2025 Feb 21];329(7):574–87. Available from: https://pubmed.ncbi.nlm.nih.gov/36809318/

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Martucci M, Russo R, Schimperna F, D'Apolito G, Panfili M, Grimaldi A, et al. Magnetic resonance imaging of primary adult brain tumors: state of the art and future perspectives. Biomedicines [Internet]. 2023 Jan 26 [cited 2025 Feb 21];11(2):364. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC9953338/

[8]

Walsh J, Othmani A, Jain M, Dev S. Using U-Net network for efficient brain tumor segmentation in MRI images. Healthc Anal [Internet]. 2022 Nov [cited 2025 Feb 21];2:100098. Available from: https://doi.org/10.1016/j.health.2022.100098

[9]

Pourmahboubi A, Arsalani Saeed N, Tabrizchi H. A brain tumor segmentation enhancement in MRI images using U-Net and transfer learning. BMC Med Imaging [Internet]. 2025 Jul 31 [cited 2025 Feb 21];25:307. Available from: https://link.springer.com/article/10.1186/s12880-025-01837-4

[10]

Abd-Ellah MK, Awad AI, Khalaf AAM, Hamed HFA. A review on brain tumor diagnosis from MRI images: practical implications, key achievements, and lessons learned. Magn Reson Imaging [Internet]. 2019 [cited 2025 Feb 21];61:300–18. Available from: https://pubmed.ncbi.nlm.nih.gov/31173851/

[11]

Rezaeikhonakdar D. AI chatbots and challenges of HIPAA compliance for AI developers and vendors. J Law Med Ethics [Internet]. 2023 Winter [cited 2025 Feb 21];51(4):988–995. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC10937180/

[12]

Havaei M, Guizard N, Chapados N, Bengio Y. HeMIS: Hetero-Modal Image Segmentation [Internet]. arXiv; 2016 Jul 18 [cited 2026 Feb 24]. Available from: https://arxiv.org/abs/1607.05194